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MSc Research Proposal

**Intelligent Decision Support for Pavement
Structural Number using Subgrade Soil
Classification**

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1 Title

Smart Decision Support for Pavement Structural Number Recommendation using Subgrade Classification

2 Introduction

Flexible pavements form the backbone of most modern road networks because of their comparatively low construction cost, ease of maintenance, and adaptability to varying traffic and environmental conditions. The structural adequacy of a flexible pavement is conventionally expressed through the *Structural Number* (SN), an empirical index introduced by the AASHTO design procedure that combines the thicknesses and material properties of individual pavement layers into a single measure of the pavement's load-carrying capacity. The design of an adequate SN is fundamentally dependent on the strength characteristics of the underlying subgrade soil, most commonly represented through the *California Bearing Ratio* (CBR).

Conventional determination of CBR and subsequent structural design requires laboratory testing, empirical correlations, and interpretation by experienced pavement engineers [3]. For practitioners without extensive geotechnical or pavement-design training, and for early-stage feasibility assessment, this process can be time-consuming and difficult to interpret. In recent years, machine learning (ML) methods have demonstrated strong potential in geotechnical and pavement engineering for capturing complex, non-linear relationships between soil properties and pavement performance indicators [4, 6]. Building on this trend, the present research proposes a *Smart Decision Support System* that predicts the Structural Number of flexible pavements from subgrade soil properties using a Random Forest (RF) regression model, converts subgrade strength into an intuitive Low/Medium/High CBR classification with a Green/Yellow/Red visual indicator, and layers a Large Language Model (LLM) interface on top of these validated outputs to provide natural-language explanation and interactive decision support for non-technical users.

3 Background of the Study

3.1 Flexible Pavement Structure

A flexible pavement is a layered structure typically consisting of a surface (wearing) course, base course, sub-base course, and subgrade. Each layer contributes a structural coefficient that, combined with layer thickness and drainage quality, determines the overall Structural Number required to withstand anticipated traffic loading over the design period.

3.2 Subgrade Soil Properties

The subgrade is the natural or compacted soil foundation upon which the pavement structure rests. Its engineering behaviour is governed by properties such as moisture content, dry unit weight, Atterberg limits (liquid limit, plastic limit, plasticity index), grain-size distribution, and compaction characteristics. These properties directly influence the subgrade's resistance to deformation under repeated loading.

3.3 California Bearing Ratio (CBR)

The CBR test is the most widely used empirical measure of subgrade strength for pavement design. It expresses the resistance of a soil to penetration relative to a standard crushed-stone material. CBR values are commonly grouped into qualitative strength categories (poor, fair, good) that inform layer-thickness design charts.

3.4 Structural Number (SN)

The Structural Number, as defined in the AASHTO 1993 design guide, is a dimensionless index summarising the overall structural capacity of a flexible pavement. It is a function of layer coefficients, layer thicknesses, and drainage coefficients, and it must be sufficient to carry the design traffic (expressed in Equivalent Single Axle Loads) over the pavement's service life without exceeding permissible serviceability loss.

3.5 Conventional Pavement Structural Design

Conventional design procedures rely on nomographs, empirical equations, or software implementations of the AASHTO guide, requiring accurate CBR or resilient-modulus input together with traffic and reliability parameters. While robust, these methods demand specialised knowledge to apply and interpret correctly.

3.6 Limitations of Conventional and Manual Approaches

Manual approaches are sensitive to interpretation errors, are not easily accessible to non-specialists, and do not scale well when rapid preliminary assessment is required across many sites or projects. They also provide limited transparency regarding which soil factors drive a given design outcome.

3.7 Machine Learning Applications in Pavement Engineering

ML techniques have been applied to a wide range of pavement problems, including pavement distress detection [1, 22], pavement roughness prediction [17], resilient modulus estimation

[18], and direct prediction of structural capacity or Structural Number from subgrade properties [8, 12].

3.8 Random Forest for Regression and Prediction

Random Forest is an ensemble learning algorithm that aggregates predictions from multiple decision trees trained on bootstrapped subsets of data and random feature subsets. It is widely favoured in geotechnical and pavement-related regression tasks for its robustness to noise, ability to model non-linear interactions, and capacity to provide feature-importance estimates [11, 13].

3.9 Decision-Support Systems

A decision-support system (DSS) combines data, models, and a user interface to assist decision-making under uncertainty. In engineering contexts, an effective DSS must not only produce a prediction but also communicate the basis and confidence of that prediction to its intended users.

3.10 Explainable and User-Friendly Pavement Recommendations

Explainable AI (XAI) methods, such as SHAP value analysis, have been increasingly applied to civil-engineering ML models to improve transparency and practitioner trust [9, 11, 21].

3.11 Potential Role of LLMs in Natural-Language Decision Support

LLMs offer the ability to translate structured numerical outputs into natural-language explanations and to answer follow-up questions interactively. Recent work has begun to explore multimodal and language-model-assisted interfaces in civil-engineering contexts [2, 20].

4 Problem Statement

Translating raw subgrade soil properties into an appropriate pavement Structural Number requires specialised knowledge of empirical design procedures, correlation charts, and engineering judgement. Non-technical stakeholders and even early-career practitioners frequently lack ready access to this expertise, leading to delays, inconsistent preliminary assessments, or reliance on generic assumptions that may not reflect site-specific subgrade conditions. Purely conventional, manual approaches provide neither rapid feedback nor transparent explanation of how a given recommendation was derived. There is therefore a need for an accessible, data-driven decision-support system that (i) reliably predicts the Structural Number from measurable subgrade properties, (ii) classifies subgrade strength into an intuitive scheme, and (iii)

communicates the reasoning behind its recommendations in plain language, while keeping the validated predictive model — rather than any generative component — as the authoritative source of the engineering estimate.

5 Research Gap

Existing studies have individually addressed several relevant sub-problems. Machine learning models, including Random Forest, Extreme Gradient Boosting, Gradient Boosting, and K-Nearest Neighbours, have been shown to predict the Structural Number of flexible pavements from subgrade properties such as dry unit weight, moisture content, weighted plasticity index, and freeze–thaw cycle count, with strong reported accuracy [12]. Separately, CBR prediction from index properties using ensemble and interpretable ML methods has been studied with explainability layers such as SHAP [11, 13, 14]. Structural-capacity estimation using ML has also been explored using pavement condition and deflection data [8]. However, the literature reveals the following gaps:

- (a) Studies that predict Structural Number using Random Forest typically stop at model evaluation and feature-importance analysis; they do not integrate the prediction into an end-to-end decision-support tool intended for non-technical users [12].
- (b) CBR classification research largely focuses on numerical prediction of CBR itself rather than translating classified subgrade strength into an actionable, colour-coded, user-facing recommendation [11].
- (c) Explainability techniques (e.g., SHAP) have been applied to related civil-engineering ML models [9, 21], but their integration with a conversational, LLM-based explanation layer specifically for pavement structural design has not been substantially explored.
- (d) No identified study combines a validated regression model for SN prediction, a rule-based CBR classification layer, and an LLM-driven natural-language interaction layer into a single, coherent decision-support architecture for flexible pavement design.

This research does not claim novelty in the individual machine-learning algorithms themselves; rather, the intended contribution lies in the integration of a validated predictive model, a transparent rule-based classification layer, and a constrained LLM explanation interface into a unified, non-technical-user-oriented decision-support system.

6 Aim of the Research

The aim of this research is to design, develop, and evaluate a machine-learning-based decision-support system that predicts the Structural Number of flexible pavements from subgrade soil

properties, classifies subgrade strength into an intuitive Low/Medium/High (Red/Yellow/Green) scheme, and provides natural-language explanation and recommendation support through an integrated LLM interface, in order to make pavement structural assessment more accessible to non-technical users while preserving engineering rigour and transparency.

7 Research Objectives

1. To identify and preprocess relevant subgrade soil and pavement-design features suitable for predicting the Structural Number.
2. To explore statistical and correlational relationships between subgrade soil properties, CBR, and Structural Number.
3. To develop a Random Forest regression model for predicting the Structural Number from subgrade properties, benchmarked against baseline regression models.
4. To evaluate the trained models using appropriate regression metrics (R^2 , RMSE, MAE) and to identify the most influential input features through feature-importance and explainability analysis.
5. To develop a transparent, rule-based CBR classification module that categorises subgrade condition into Low/Medium/High classes with Red/Yellow/Green indicators.
6. To design a decision-support layer that combines the predicted Structural Number, CBR classification, and key influencing factors into an understandable recommendation for non-technical users.
7. To integrate a Large Language Model API as an auxiliary natural-language explanation and interaction component, without allowing it to override the validated Random Forest prediction.

8 Research Questions

- RQ1. Which subgrade soil properties have the greatest influence on the predicted Structural Number?
- RQ2. How accurately can a Random Forest model predict Structural Number compared to baseline regression models?
- RQ3. How can CBR values and associated soil characteristics be classified into meaningful, defensible subgrade-condition categories (Low/Medium/High)?

- RQ4. Can the proposed system provide understandable and actionable pavement-structure recommendations to non-technical users?
- RQ5. How can an integrated LLM component improve explanation quality and user interaction without compromising the authority of the validated machine-learning prediction?

9 Research Hypotheses

- **H1:** A Random Forest regression model can predict the flexible pavement Structural Number from subgrade soil properties with higher accuracy (lower RMSE/MAE, higher R^2) than a simple linear-regression baseline.
- **H2:** Subgrade strength-related features (e.g., CBR-correlated indices, plasticity index, moisture content) will rank among the most important predictors of Structural Number in feature-importance analysis.
- **H3:** A rule-based CBR classification layer, grounded in established design thresholds, will produce subgrade-condition labels that are consistent with the trends captured by the Random Forest model's predictions.

10 Literature Review

10.1 Flexible Pavement Design and Subgrade Characterisation

Empirical pavement design methods, most notably the AASHTO 1993 guide, rely on subgrade strength parameters such as CBR or resilient modulus to determine the required Structural Number [3]. Subgrade characterisation datasets, such as regional soil-property collections, provide the geotechnical basis for such design procedures [5].

10.2 CBR and Soil Strength Prediction

Several studies have applied machine learning to predict CBR directly from index properties. Korkmaz Tan and Ordu applied ensemble methods combined with SHAP analysis for interpretable CBR prediction [11]. Marathe and Sheshadri predicted CBR for lateritic–lithomargic subgrades using major index properties [13]. Wimalasena and Gallage examined CBR prediction for selected subgrade materials using statistical and machine-learning approaches [16], while Elhassasn et al. proposed an optimised deep-learning framework for CBR prediction [14].

10.3 Structural Number and Structural Capacity Prediction

Ziar evaluated Random Forest, Extreme Gradient Boosting, Gradient Boosting, and K-Nearest Neighbour regressors for predicting the Structural Number of flexible pavements using dry unit weight, moisture content, weighted plasticity index, and freeze–thaw cycle count as inputs, reporting strong predictive performance for the tree-based ensemble methods and highlighting subgrade strength-related variables among the dominant predictive features [12]. Karballaezadeh et al. used machine-learning techniques to estimate flexible pavement structural capacity from pavement condition and deflection-related data [8]. Related work on subgrade resilient modulus under freeze–thaw and moisture variation using gene-expression programming and neural networks further supports the relevance of environmental and moisture-related variables in structural prediction tasks [18].

10.4 Machine Learning in Broader Pavement and Geotechnical Engineering

Machine learning has additionally been applied to pavement distress and crack detection [1, 20, 22], road-surface classification using inertial measurement unit data [7, 19], pavement roughness prediction [17], and structural health monitoring [6]. Beyond pavements, similar ML approaches have supported soil classification and agricultural decision support [2, 15], concrete-strength and mix-design optimisation [9, 21], and broader geotechnical applications [4], illustrating a wider trend toward data-driven decision support across civil engineering. GIS-based subgrade bearing-capacity zonation further demonstrates the value of spatially aware, data-driven subgrade assessment [10].

10.5 Explainability and Rule-Based Decision Systems

Explainable AI techniques such as SHAP have been used to interpret ML predictions in concrete-strength modelling [21], fly-ash concrete design optimisation [9], and CBR prediction [11], supporting the case for combining predictive accuracy with interpretability in engineering decision-support tools. Rule-based classification layers remain valuable where transparent, standards-traceable thresholds are required, complementing black-box predictive models rather than replacing them.

10.6 LLMs in Engineering Decision Support

Emerging work has begun exploring multimodal and language-model-based interfaces for engineering tasks, including vision-language models for pavement crack description [20] and LLM-assisted recommendation layers in agricultural decision-support systems [2]. This litera-

ture motivates, but has not yet substantially addressed, the specific integration of an LLM explanation layer with a validated Structural Number prediction and CBR classification pipeline.

11 Proposed Methodology

11.1 Stage 1: Dataset Acquisition

The primary dataset will consist of subgrade soil and pavement-related records containing, where available: CBR, AASHTO/USCS soil classification, natural moisture content, maximum dry density, optimum moisture content, Atterberg limits (liquid limit, plastic limit, plasticity index), grain-size distribution and fines content (percentage passing No. 200 sieve), compaction characteristics, freeze–thaw cycle count where applicable, and, where available, corresponding traffic/loading information and an existing or back-calculated Structural Number as the target variable. Candidate sources include published subgrade-characterisation datasets [5] and datasets associated with recent Structural Number prediction studies [12]. Variables that are directly available in the acquired dataset will be clearly distinguished from variables that would require supplementary collection, correlation, or estimation.

11.2 Stage 2: Data Preprocessing

Preprocessing will include missing-value handling (imputation or exclusion, depending on missingness pattern), outlier detection using standard statistical methods (e.g., interquartile range or z-score screening), verification of data consistency and unit standardisation, categorical feature encoding, and feature scaling where required by the chosen algorithms. Exploratory data analysis (distribution plots, box plots) and correlation analysis (Pearson/Spearman) will guide feature-selection decisions. The dataset will be partitioned into training, validation, and test subsets (e.g., an 80/20 or 70/15/15 split, to be finalised based on dataset size), with care taken to avoid data leakage — particularly to exclude any variable that is definitionally or near-definitionally derived from the Structural Number itself.

11.3 Stage 3: Random Forest Model Development

A Random Forest regression model will be developed as the primary predictor of Structural Number. Model development will include hyperparameter tuning (e.g., number of trees, maximum depth, minimum samples per leaf) via grid or randomised search with k-fold cross-validation, extraction of built-in feature-importance scores, and steps to mitigate overfitting (cross-validation, regularisation of tree depth, out-of-bag error monitoring). The Random Forest model will be benchmarked against baseline models, namely Linear Regression, Decision Tree Regression, and Gradient Boosting Regression (or Extreme Gradient Boosting), following

the comparative approach used in related Structural Number prediction work [12]. The comparison is intended to justify the final model choice on accuracy, robustness, and interpretability grounds rather than to report benchmark numbers in isolation.

11.4 Stage 4: Model Evaluation

Model performance will be assessed using the coefficient of determination (R^2), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and, where mathematically appropriate (i.e., where target values are strictly positive and away from zero), Mean Absolute Percentage Error (MAPE). R^2 indicates the proportion of variance in Structural Number explained by the model; RMSE penalises larger errors more heavily and is useful for detecting poor performance on outlying cases; MAE gives an easily interpretable average error magnitude; MAPE, where applicable, expresses error relative to the magnitude of the target value.

11.5 Stage 5: CBR-Based Classification

A transparent, rule-based classification module will convert CBR and related subgrade indicators into three categories: Low/Red (poor subgrade condition), Medium/Yellow (moderate subgrade condition), and High/Green (good subgrade condition). Thresholds will not be selected arbitrarily; they will be derived from established pavement-design guidance and cross-referenced against the distribution of CBR values in the acquired dataset and against thresholds reported in related CBR-classification literature [11, 13], with the final thresholds and their justification documented explicitly in the dissertation.

11.6 Stage 6: Decision-Support Layer

The decision-support layer will combine the Random Forest-predicted Structural Number, the CBR classification, the values of the most influential soil properties, and, where feasible, an indication of prediction uncertainty (e.g., variance across the Random Forest's constituent trees) into a single, non-technical-user-facing output. A representative output will include: the CBR condition (Green/Yellow/Red), the predicted Structural Number, a recommended pavement structural category, a short list of key influencing factors, and an explicit disclaimer indicating that professional engineering validation is required before the recommendation is used for final design.

11.7 Stage 7: LLM Integration

An LLM will be integrated as a natural-language explanation and interaction layer, accessed via an API in a manner consistent with an existing modular integration approach [15]. The LLM will be prompted to explain the Random Forest prediction and CBR classification in

plain language, describe why the subgrade received a particular classification, summarise the influence of key soil properties, answer user follow-up questions, and provide general (non-authoritative) recommendations regarding pavement-design terminology and next steps. The LLM will not be permitted to independently generate or override the Structural Number; its inputs will be restricted to the structured outputs of the Random Forest model and the rule-based classifier. Hallucination-mitigation strategies will include retrieval-augmented grounding against a curated reference set where feasible, tightly controlled and templated prompts, requiring the LLM to cite only the system's own structured outputs, and mandatory inclusion of engineering-validation disclaimers in every generated response.

12 Proposed System Architecture

Figure 1 illustrates the conceptual system architecture.

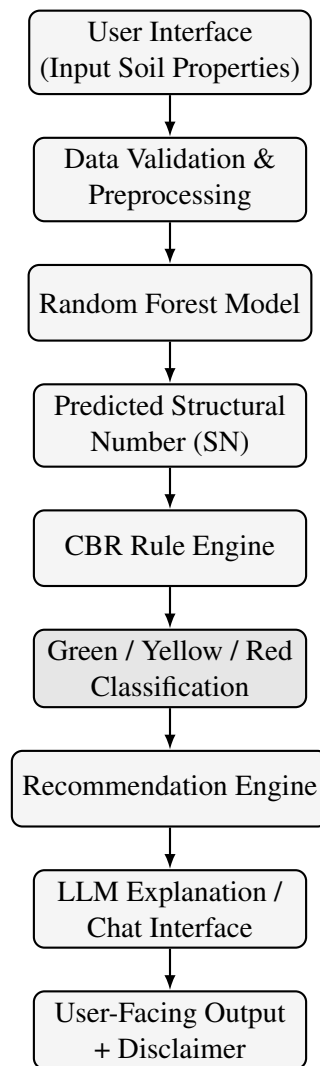


Figure 1: Conceptual architecture of the proposed Smart Decision Support System.

User Interface collects subgrade soil property inputs from the user. **Data Validation & Pre-**

processing checks input ranges/units and applies the same transformations used in model training. The **Random Forest Model** produces the predicted Structural Number, which is passed to the **CBR Rule Engine** for classification into Green/Yellow/Red subgrade condition. The **Recommendation Engine** compiles the SN, classification, and key influencing factors into a structured recommendation, which the **LLM Explanation/Chat Interface** translates into natural language and uses to answer follow-up questions, before the final **User-Facing Output**, including a mandatory engineering-validation disclaimer, is returned.

13 Expected Results

Without pre-empting the outcome of experimentation, the research is expected to yield: a trained and validated Random Forest model capable of predicting flexible pavement Structural Number from subgrade properties with accuracy comparable to that reported for similar models in the literature [12]; a ranked identification of the subgrade features most influential in determining Structural Number; a transparent, standards-referenced CBR classification mechanism; a working prototype decision-support interface suitable for use by non-technical stakeholders; and a demonstration of LLM-generated natural-language explanations that remain consistent with, and do not contradict, the underlying model outputs.

14 Expected Research Contributions

- **Scientific contribution:** Comparative evidence on the relative performance of Random Forest and baseline regression models for Structural Number prediction on the assembled dataset.
- **Methodological contribution:** A documented, standards-referenced procedure for deriving CBR classification thresholds and integrating them with a machine-learning prediction pipeline.
- **Practical/engineering contribution:** A prototype tool supporting rapid, preliminary pavement structural assessment for practitioners and non-specialist stakeholders.
- **Decision-support contribution:** A three-layer architecture (prediction, rule-based classification, and natural-language explanation) that keeps a validated ML model as the authoritative engineering component.
- **Human-computer interaction/LLM contribution:** An applied demonstration of constrained, grounded LLM usage as an explanation layer within an engineering decision-support context, together with an assessment of associated risks and mitigation strategies.

15 Scope of the Research

This research is limited to **flexible** pavements; rigid pavement design is outside its scope. The study will use whichever subgrade-characterisation and Structural Number datasets can be acquired or compiled within the MSc timeframe, and results will be interpreted with respect to the soil types and geographic regions represented in that dataset. The CBR classification thresholds will be derived from the specific dataset and referenced literature/standards used in this study and may require recalibration for other regions. The LLM component is scoped strictly as an explanation and interaction layer and is explicitly not scoped to perform independent engineering design.

16 Research Limitations

Anticipated limitations include: potential dataset bias toward the soil types, climates, and testing standards represented in the source data; limited generalisability of the trained model beyond the training data's geographic and geotechnical range; measurement uncertainty inherent in laboratory-derived CBR and Atterberg-limit values; a potentially limited number of training samples relative to the dimensionality of engineering variability; possible distribution shift when applied to unseen sites; inherent interpretability limits of ensemble tree models despite feature-importance analysis; and the risk of LLM hallucination or overconfident phrasing, which will be mitigated but not entirely eliminated by the grounding strategies described in Stage 7 of the methodology.

17 Ethical and Safety Considerations

The proposed system is intended strictly as a decision-support and preliminary-assessment tool. It is not designed, and must not be represented, as a replacement for professional pavement engineering design or for the judgement of a licensed engineer. All outputs will carry an explicit disclaimer indicating that final structural design decisions require validation by a qualified engineer. Responsible use of AI in engineering will be discussed, including the risks of over-reliance on automated recommendations, the importance of maintaining a human-in-the-loop for consequential decisions, and the need for clear communication of model confidence and limitations to end users.

18 Validation Strategy

System validation will proceed at three levels:

1. **Machine-learning validation:** Predictive performance of the Random Forest model (and baselines) will be assessed on a held-out test set using R^2 , RMSE, and MAE, together with cross-validation to assess generalisation stability.
2. **Rule-based validation:** The CBR classification thresholds will be checked for consistency against established pavement-design guidance and against independently reported CBR-classification schemes in the literature.
3. **System/user validation:** A small-scale usability evaluation with a limited number of participants (e.g., students, junior engineers, or non-technical stakeholders) will assess task completion, comprehension of the recommendation, and overall satisfaction, potentially using a System Usability Scale (SUS) questionnaire.

19 Research Workflow

The overall workflow proceeds as follows: (1) literature review and refinement of the research problem; (2) dataset acquisition and preprocessing; (3) exploratory data analysis and correlation study; (4) development and hyperparameter tuning of the Random Forest and baseline models; (5) model evaluation and feature-importance/explainability analysis; (6) development of the CBR rule-based classification module; (7) development of the decision-support and recommendation layer; (8) integration of the LLM explanation interface; (9) system-level and usability testing; and (10) documentation, analysis, and thesis writing.

20 Proposed Timeline

Period	Activities
Months 1–2	Literature review; refinement of problem statement, objectives, and research questions.
Months 2–3	Dataset acquisition and compilation; initial data quality assessment.
Months 3–4	Data preprocessing, exploratory data analysis, correlation and feature-selection study.
Months 4–6	Random Forest model development, hyperparameter tuning, and baseline model comparison.
Month 6	Model evaluation (R^2 , RMSE, MAE) and feature-importance/explainability analysis.
Months 7–8	Development of CBR rule-based classification module and threshold justification.

Period	Activities
Month 8	Development of the decision-support/recommendation layer.
Months 9–10	Integration of LLM API, prompt design, and hallucination-mitigation measures.
Month 10	System testing, integration testing, and small-scale usability validation.
Months 11–12	Thesis writing, revision, and final submission.

21 Required Tools and Technologies

Tool/Technology	Justification
Python	Primary programming language; extensive ecosystem for data science and machine learning.
Pandas, NumPy	Data manipulation, cleaning, and numerical computation.
Scikit-learn	Implementation of Random Forest, baseline regressors, cross-validation, and evaluation metrics.
Matplotlib / Seaborn	Visualisation of exploratory data analysis and model performance.
SHAP	Model explainability and feature-importance/contribution analysis.
Flask / FastAPI / Streamlit	Deployment of the decision-support interface for end-user interaction.
Relational or lightweight database (e.g., SQLite/PostgreSQL)	Storage of input records, predictions, and interaction logs, if required.
LLM API	Provision of the natural-language explanation and interaction layer.

22 Data and Model Governance

Data governance will address data quality checks at ingestion, versioning of both datasets and trained models to support reproducibility, and clear documentation of preprocessing steps. Model governance will include recording of hyperparameters and training configurations for each experiment. API security measures will include secure storage of credentials and rate-limiting where applicable. Privacy will be addressed by avoiding collection of any personally

identifiable information beyond what is strictly necessary for system usability testing, and by anonymising participant data in the validation study. Logging will capture system inputs and outputs (excluding sensitive personal data) to support debugging and monitoring. Prompt security measures will guard against prompt injection affecting the LLM layer. Ongoing model monitoring and periodic retraining considerations will be discussed as part of the system's long-term maintainability, and all experiments will be documented to support reproducibility.

23 Supporting Tables

23.1 Objectives, Research Questions, Methods, and Expected Outputs

Objective	Related RQ(s)	Method	Expected Output
Obj. 1: Feature identification & preprocessing	RQ1	Data cleaning, correlation analysis, feature selection	Cleaned, feature-selected dataset
Obj. 2: Explore relationships	RQ1	Exploratory data analysis, correlation study	Documented relationships between soil properties, CBR, and SN
Obj. 3: Random Forest model development	RQ2	RF regression, hyperparameter tuning, baseline comparison	Trained and tuned Random Forest model
Obj. 4: Model evaluation & explainability	RQ1, RQ2	R^2 , RMSE, MAE; feature importance / SHAP	Performance report and feature-importance ranking
Obj. 5: CBR classification	RQ3	Rule-based thresholding referenced to design standards	Low/Medium/High (Red/Yellow/Green) classifier
Obj. 6: Decision-support layer	RQ4	Integration of SN, classification, and key factors	User-facing structured recommendation
Obj. 7: LLM integration	RQ5	API integration, prompt design, grounding, disclaimers	Interactive natural-language explanation interface

23.2 Proposed Dataset Features and Expected Roles

Feature	Role	Notes
CBR	Predictor / classification basis	Central subgrade-strength indicator
Soil classification (AASHTO/USCS)	Predictor	Categorical; may require encoding
Moisture content	Predictor	Reflects field/compaction condition
Maximum dry density	Predictor	From standard compaction test
Optimum moisture content	Predictor	From standard compaction test
Liquid limit, plastic limit, plasticity index	Predictor	Atterberg limits; plasticity behaviour
Grain-size distribution / fines content	Predictor	% passing No. 200 sieve, etc.
Freeze–thaw cycle count	Predictor (where available)	Environmental factor influencing strength
Traffic/loading data	Contextual predictor (if available)	May supplement subgrade-only inputs
Structural Number (SN)	Target variable	Predicted output of the Random Forest model

23.3 Preliminary List of Evaluation Metrics

Metric	Purpose
R^2	Proportion of variance in Structural Number explained by the model.
RMSE	Penalises larger prediction errors; sensitive to outliers.
MAE	Average magnitude of prediction error; easily interpretable.
MAPE (where applicable)	Error expressed as a percentage of the true value; used only where targets are safely away from zero.

23.4 Preliminary List of Relevant Recent Research Papers and References

A preliminary, non-exhaustive selection of directly relevant recent papers includes: Ziar's Random Forest, XGBoost, Gradient Boosting, and KNN comparison for Structural Number prediction [12]; Korkmaz Tan and Ordu's interpretable ensemble CBR prediction with SHAP [11]; Marathe and Sheshadri's CBR prediction for lateritic–lithomargic subgrades [13]; Karballaezadeh et al.'s ML-based structural-capacity estimation [8]; Elhassasn et al.'s optimised deep-learning CBR prediction framework [14]; and Baghbani et al.'s state-of-the-art review of AI in geotechnical engineering [4]. The full reference list is provided in Section 24.

24 References

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